

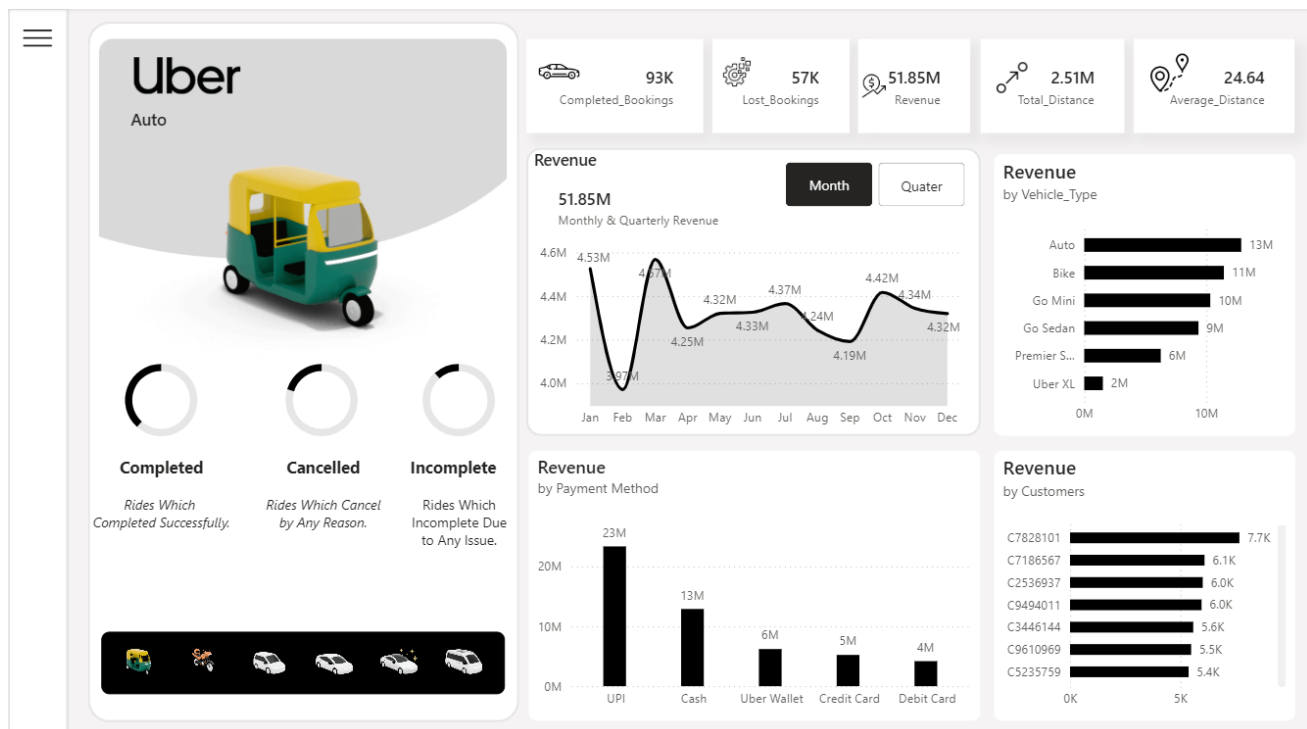
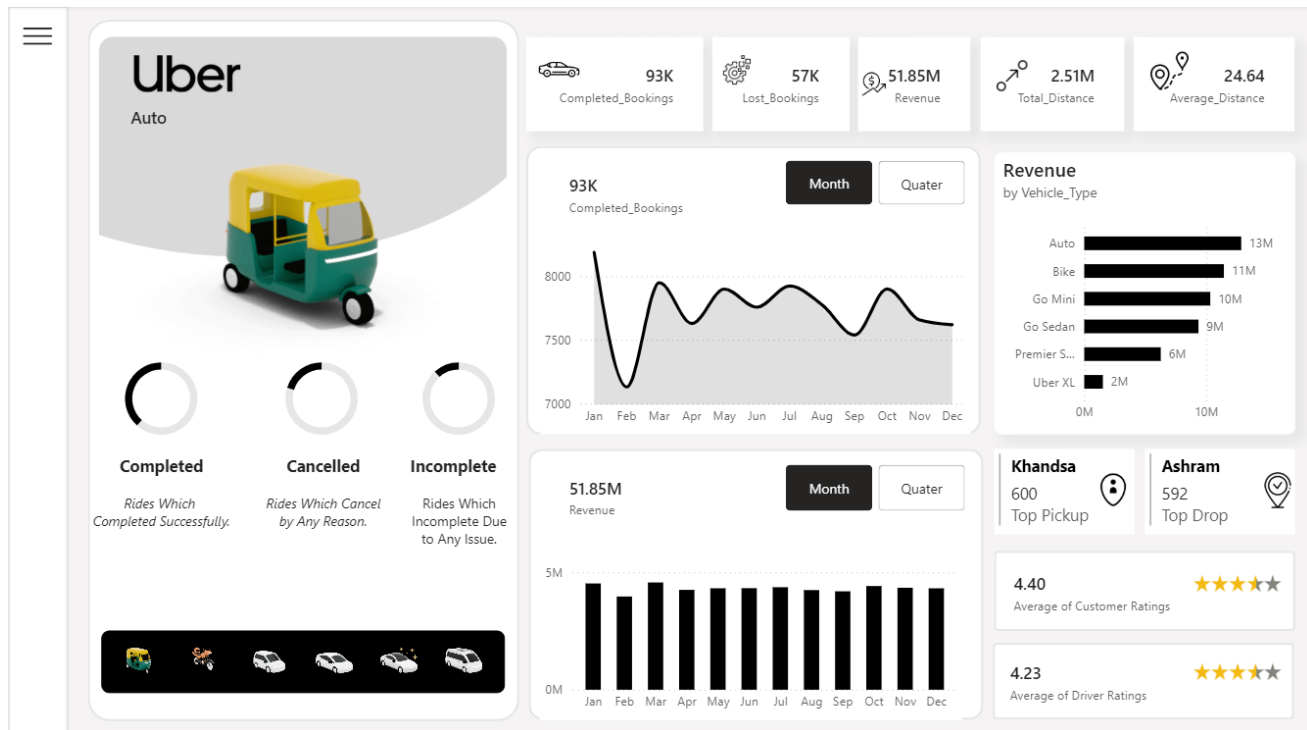
UBER RIDE ANALYTICS

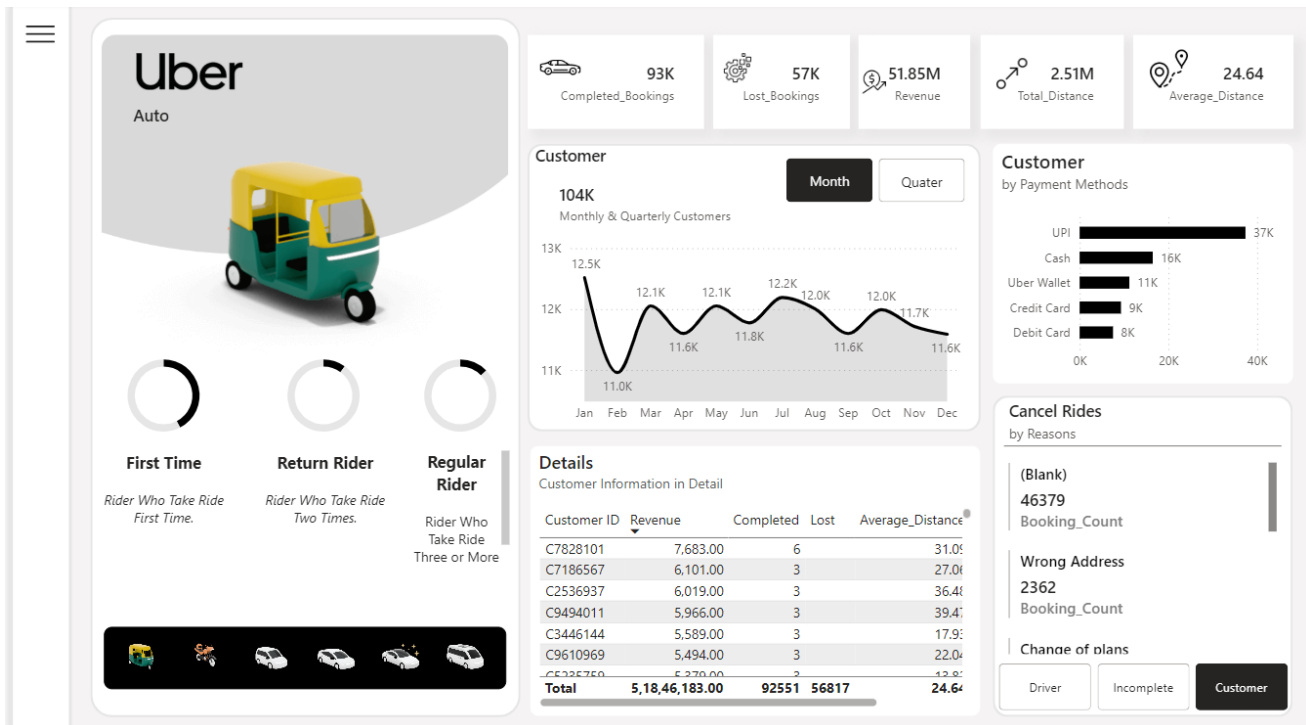
End-to-End Business Intelligence & Power BI Dashboard Documentation

Dataset: 150,000 ride-booking records | Analysis period: 2025

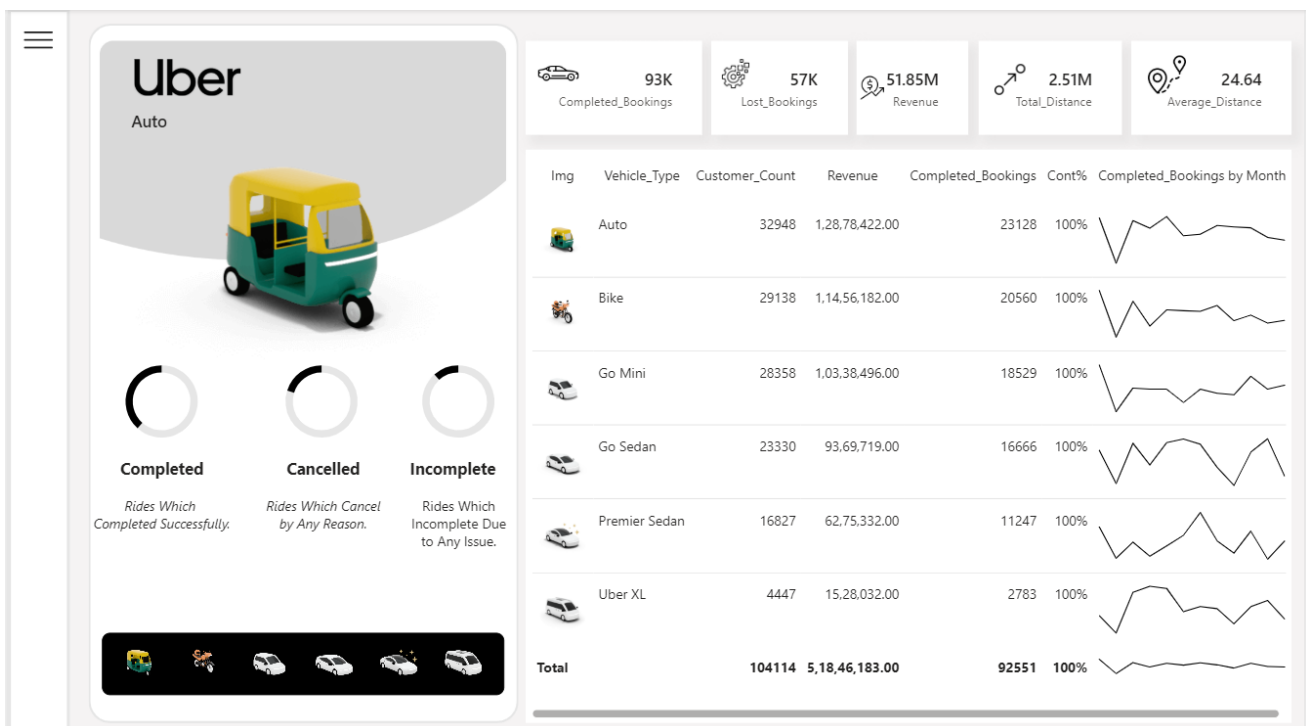


Home / Landing Page





Rider – Customer Segmentation and Cancellation Analysis



Vehicle – Vehicle-Type Performance Detail

1. Executive Summary

This project analyzes 150,000 Uber ride-booking records covering 2025. The Power BI solution converts booking, revenue, vehicle, customer, payment, distance, rating and cancellation data into an interactive business dashboard.

The dataset contains 93,000 completed bookings (62.0%), while 57,000 bookings (38.0%) did not result in a completed trip. The dashboard reports ₹51.85M of Booking Value, 2.51M total ride distance, 104,114 unique customers, a 4.40/5 customer rating and a 4.23/5 driver rating.

The strongest commercial signals are the dominance of Auto in revenue, UPI as the leading payment method, relatively stable month-to-month revenue, and a significant 38% booking-loss rate. The largest operational opportunity is therefore not only revenue growth but also conversion improvement: reducing driver/customer cancellations, no-driver cases and incomplete rides.

2. Business Problem

A ride-hailing business generates large volumes of transactional data, but raw booking records do not directly answer management questions. The business needs a consolidated analytical view to understand demand, revenue performance, vehicle mix, customer behavior and service failures.

Key business questions

- How many bookings are completed versus lost, cancelled or incomplete?
- Which months generate the highest and lowest revenue and completed bookings?
- Which vehicle categories contribute the most revenue and completed trips?
- Which payment methods are most important for revenue?
- Which pickup and drop locations have the highest completed-trip activity?
- Who are the highest-value customers?
- What are the major reasons for customer and driver cancellations?
- How large is the first-time, returning and regular customer base?
- How are customer and driver ratings performing?
- Where should operations, customer experience and retention efforts be focused?

3. Project Objectives

- Build a single source of truth for key Uber ride KPIs.
- Monitor booking conversion and identify booking leakage.
- Analyze revenue trends by month, vehicle type, payment method and customer.
- Understand rider behavior and identify retention opportunities.
- Identify operational pain points through cancellation and incomplete-ride reasons.
- Provide an interactive dashboard with slicers, navigation and drill-oriented detail views.
- Translate descriptive analytics into practical business recommendations.

4. Dataset Overview

Source file: uber.xlsx. The primary sheet (Sheet1) contains 150,000 records and 19 columns. The data spans 2025-01-01 to 2025-12-30. Supporting sheets include vehicle-image/reference data used by the dashboard.

Metric	Value
Rows	150,000
Columns	19
Date range	2025-01-01 to 2025-12-30

Unique customers	104,114
Unique booking IDs	148,767
Vehicle types	7
Pickup locations	176
Drop locations	176

Data dictionary

Column	Meaning	Business use
Date	Booking date	Time-series analysis
Time	Booking time	Potential time-of-day analysis
Booking ID	Booking identifier	Booking-level tracking
Booking Status	Completed/cancelled/no-driver/incomplete	Funnel and operational performance
Customer ID	Customer identifier	Customer frequency and value
Vehicle Type	Selected ride category	Vehicle mix analysis
Pickup Location	Origin location	Demand and hotspot analysis
Drop Location	Destination location	Destination analysis
Cancelled Rides by Customer	Customer cancellation flag	Customer-side cancellation measurement
Reason for cancelling by Customer	Customer cancellation reason	Root-cause analysis
Cancelled Rides by Driver	Driver cancellation flag	Driver-side cancellation measurement
Driver Cancellation Reason	Driver cancellation reason	Operational root-cause analysis
Incomplete Rides	Incomplete-trip flag	Service failure measurement
Incomplete Rides Reason	Incomplete-trip reason	Root-cause analysis
Booking Value	Booking amount	Revenue / booking value analysis
Ride Distance	Trip distance	Distance and utilization analysis
Driver Ratings	Customer's driver rating	Driver experience KPI
Customer Rating	Driver/customer-facing rating field	Customer experience KPI
Payment Method	UPI/Cash/Wallet/Card	Payment mix analysis

5. Data Preparation & Analytical Methodology

- Imported the Excel dataset into Power BI.
- Validated data types for dates, numeric measures and categorical fields.
- Reviewed missing values and used the booking status to interpret sparse fields correctly.
- Created/used calendar/date-axis structures to support monthly and quarterly analysis.
- Created KPI measures for completed bookings, lost bookings, booking value/revenue, total distance, average distance and customer count.
- Built analytical views for booking trends, revenue trends, vehicle performance, payment methods, customers, rider behavior and cancellation reasons.
- Added interactive navigation and vehicle selectors to make the report behave like a multi-page analytical application.

Important analytical definition

The dashboard's reported Revenue/Booking Value total is ₹51.85M and is calculated from all non-blank Booking Value records. Because the source contains Booking Value for 9,000 incomplete rides, this figure should be interpreted as reported Booking Value rather than strictly realized revenue. Completed-booking Booking Value is ₹47.26M. For a production finance dashboard, a separate Realized Revenue measure filtered to Completed bookings is recommended.

6. KPI Summary

KPI	Result	Interpretation
Total bookings	150,000	All records
Completed bookings	93,000	0.6% of bookings
Lost bookings	57,000	0.4% of bookings

Reported Booking Value	₹51.85M	All non-blank Booking Value
Completed-booking value	₹47.26M	Completed only
Total distance	2.51M	All non-blank Ride Distance
Average distance	24.64	All non-blank Ride Distance
Unique customers	104,114	Distinct Customer IDs
Customer rating	4.40/5	Average
Driver rating	4.23/5	Average

7. Booking Status Analysis

Booking status	Bookings	Share
Completed	93,000	62.0%
Cancelled by Driver	27,000	18.0%
Cancelled by Customer	10,500	7.0%
No Driver Found	10,500	7.0%
Incomplete	9,000	6.0%

Finding: only 62% of bookings reach completion. Driver cancellations account for 18%, customer cancellations 7%, no-driver cases 7% and incomplete rides 6%. In other words, the business loses roughly 38 out of every 100 booking attempts before a successful completion.

8. Monthly Performance Findings

Month	Bookings	Completed	Booking Value	Completion rate
Jan	13,245	8,189	₹4.53M	61.83%
Feb	11,543	7,130	₹3.97M	61.77%
Mar	12,719	7,954	₹4.57M	62.54%
Apr	12,199	7,632	₹4.25M	62.56%
May	12,778	7,905	₹4.32M	61.86%
Jun	12,440	7,757	₹4.33M	62.36%
Jul	12,897	7,926	₹4.37M	61.46%
Aug	12,636	7,780	₹4.24M	61.57%
Sep	12,248	7,542	₹4.19M	61.58%
Oct	12,651	7,905	₹4.42M	62.49%
Nov	12,394	7,659	₹4.34M	61.80%
Dec	12,250	7,621	₹4.32M	62.21%

Finding: monthly Booking Value is relatively stable, ranging from ₹3.97M to ₹4.57M. Mar is the strongest revenue month (₹4.57M), while Feb is the weakest (₹3.97M). Jan has the highest completed-booking volume (8,189), while Feb is lowest (7,130). This indicates a fairly consistent demand base rather than extreme seasonality.

9. Vehicle-Type Performance

Vehicle	Bookings	Completed	Booking Value	Revenue share	Completion rate
Auto	37,419	23,155	₹12.88M	24.84%	61.88%
Go Mini	29,806	18,549	₹10.34M	19.94%	62.23%
Go Sedan	27,141	16,676	₹9.37M	18.07%	61.44%
Bike	22,517	14,034	₹7.84M	15.12%	62.33%
Premier Sedan	18,111	11,252	₹6.28M	12.10%	62.13%
eBike	10,557	6,551	₹3.62M	6.98%	62.05%
Uber XL	4,449	2,783	₹1.53M	2.95%	62.55%

Finding: Auto is the leading vehicle category, contributing ₹12.88M (24.84% of reported Booking Value) and 23,155 completed bookings. Go Mini and Go Sedan follow. Completion rates are tightly clustered around 61–63%, suggesting that the booking-loss problem is broad-based rather than isolated to a single vehicle category.

10. Payment Method Analysis

Payment method	Bookings	Booking Value	Revenue share
UPI	45,909	₹23.35M	45.03%
Cash	25,367	₹12.90M	24.87%
Uber Wallet	12,276	₹6.20M	11.96%
Credit Card	10,209	₹5.22M	10.08%
Debit Card	8,239	₹4.18M	8.06%

Finding: UPI is the dominant payment method, generating ₹23.35M or 45.0% of reported Booking Value. Cash is second. This makes UPI a strategically important payment channel and a useful area for payment reliability, promotion and customer-conversion initiatives.

11. Cancellation & Incomplete-Ride Root Causes

Customer cancellations

Reason	Count	Share of customer cancellations
Wrong Address	2,362	22.5%
Change of plans	2,353	22.4%
Driver is not moving towards pickup location	2,335	22.2%
Driver asked to cancel	2,295	21.9%
AC is not working	1,155	11.0%

Driver cancellations

Reason	Count	Share of driver cancellations
Customer related issue	6,837	25.3%
The customer was coughing/sick	6,751	25.0%
Personal & Car related issues	6,726	24.9%
More than permitted people in there	6,686	24.8%

Incomplete rides

Reason	Count	Share of incomplete rides
Customer Demand	3,040	33.8%
Vehicle Breakdown	3,012	33.5%
Other Issue	2,948	32.8%

Finding: customer-cancellation reasons are fairly evenly distributed, with Wrong Address and Change of Plans slightly leading. Driver cancellations are also distributed almost evenly across the four recorded reasons. Incomplete rides are similarly balanced among Customer Demand, Vehicle Breakdown and Other Issue. Because no single reason dominates, the best response is a combination of process improvements rather than a single fix.

12. Location Findings

The dashboard's Top Pickup and Top Drop cards are based on completed rides.

Rank	Top pickup	Completed pickups	Top drop	Completed drops
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1	Khandsa	600	Ashram	592
2	Barakhamba Road	594	Preet Vihar	589
3	Subhash Chowk	582	Sultanpur	584
4	Madipur	579	Noida Extension	579
5	Mehrauli	574	Dwarka Mor	574
6	Kanhaiya Nagar	572	Narsinghpur	574
7	Ashok Park Main	568	Sarai Kale Khan	573
8	Badarpur	567	Lajpat Nagar	572
9	Dwarka Sector 21	565	Cyber Hub	571
10	Lok Kalyan Marg	564	Kashmere Gate ISBT	570

Finding: Khandsa is the leading completed-ride pickup location with 600 rides, while Ashram is the leading completed-ride drop location with 592 rides. These locations can be used to prioritize driver availability, pickup experience and demand forecasting.

13. Customer & Rider Findings

The dataset contains 104,114 unique customers. Among customers with at least one completed ride, 64,575 customers have completed a ride. Using completed-booking frequency as a simple portfolio segmentation, 47,581 are first-time customers (1 completed ride), 5,650 are return riders (2 completed rides), and 11,344 are regular riders (3+ completed rides).

The Rider page is designed around first-time, return and regular-rider behavior, helping management move beyond booking volume toward retention. The main business opportunity is to convert first-time riders into repeat users.

Top customers by completed-booking value

Customer ID	Booking Value	Completed rides	Avg. distance
C7828101	₹7,683	6	31.09
C7186567	₹6,101	3	27.06
C2536937	₹6,019	3	36.48
C9494011	₹5,966	3	39.47
C3446144	₹5,589	3	17.93
C9610969	₹5,494	3	22.04
C5235759	₹5,379	3	13.82
C1578062	₹5,368	3	23.84
C3780888	₹5,323	3	18.14
C3791237	₹5,292	3	15.58

The highest-value customer in the dataset is C7828101, with ₹7,683 across 6 completed rides.

14. Customer & Driver Experience

Metric	Average	Interpretation
Customer rating	4.40/5	Strong overall customer experience signal
Driver rating	4.23/5	Positive driver-side service perception

The ratings are healthy overall, but they should be monitored alongside cancellation and incomplete-trip metrics. A good average rating does not eliminate operational friction if a large proportion of booking attempts still fail.

15. Dashboard Architecture & Page-by-Page Explanation

Page	Purpose
Home	Landing/navigation page with Uber branding, project title and navigation buttons. It establishes the purpose of the report and provides a clean entry point.

Overview	Executive performance page with headline KPIs, completed-booking trend, monthly revenue, revenue by vehicle, top pickup/drop locations and customer/driver ratings.
Revenue	Revenue-focused page showing monthly/quarterly Booking Value, vehicle revenue, payment-method revenue and top customers.
Rider	Customer-focused page showing customer trend, payment behavior, rider segmentation, detailed customer table and cancellation reasons.
Vehicle	Vehicle-level detail page showing vehicle images, customer count, revenue, completed bookings, contribution percentage and monthly booking trends.

16. Consolidated Business Findings

- Booking conversion is the biggest operational issue: only 62% of booking attempts are completed, leaving 57K lost bookings.
- Driver cancellations are the largest single loss category at 27K (18% of all bookings).
- Revenue is relatively stable across the year; Mar is strongest and Feb is weakest, so growth should not rely only on seasonality.
- Auto is the top revenue vehicle and the largest completed-booking contributor.
- UPI is the most important payment channel, accounting for about 45% of reported Booking Value.
- Khandsa and Ashram are the top completed-ride pickup and drop locations respectively.
- Customer and driver ratings are strong at 4.40 and 4.23, but operational losses remain high.
- Cancellation and incomplete-ride reasons are relatively evenly distributed, indicating a multi-factor operational problem.
- A significant first-time customer base creates an opportunity for retention campaigns and second-ride conversion.
- Reported Booking Value includes incomplete rides; a finance-ready dashboard should distinguish booked value from realized revenue.

17. Business Recommendations

- 1. Reduce driver cancellations:** Investigate driver-side cancellation patterns, improve driver-job matching, use clearer pickup information and review cases involving customer-related and vehicle/personal issues.
- 2. Improve driver availability:** No-driver cases represent 7% of all bookings. Use location-level demand forecasting and supply rebalancing around high-volume pickup areas.
- 3. Improve pickup accuracy:** Wrong Address is the leading customer cancellation reason. Improve address validation, map-pin accuracy and pickup instructions.
- 4. Target first-to-second ride conversion:** Create post-trip offers, reminders and loyalty incentives specifically for first-time riders.
- 5. Protect the UPI funnel:** Since UPI is the leading payment method, monitor failed UPI transactions and ensure payment retries/fallback options are smooth.
- 6. Prioritize high-demand locations:** Use top pickup/drop hotspots to plan driver positioning, incentives and operational staffing.
- 7. Separate realized revenue:** Create a Completed Revenue measure that only includes Completed bookings, while retaining Booking Value as a separate commercial metric.
- 8. Monitor vehicle economics:** Auto, Go Mini and Go Sedan drive most value. Track revenue per completed ride, distance, cancellation rate and contribution margin by vehicle category.

9. Build a retention dashboard: Track first-to-second ride conversion, repeat rate, customer lifetime value and cohort retention by acquisition month.

10. Add operational SLAs: If trip timestamps become available, add driver acceptance time, pickup wait time, trip duration and cancellation-before-pickup metrics.

18. Conclusion

The Uber Ride Analytics project demonstrates how transactional ride data can be transformed into a management-ready Business Intelligence solution. The dashboard provides a clear view of demand, revenue, customer behavior, vehicle mix and operational losses. The most important strategic takeaway is that improving booking completion and customer retention can potentially create more value than focusing only on top-line booking volume. With stronger financial definitions and additional operational data, this dashboard can be extended into a production-grade mobility analytics solution.